



The Legislative Paradigm of China's Biosafety Law and Its Positive Significance for the SARS-CoV-2 Prevention and Control

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1. INTRODUCTION

BIOSAFETY HAS BECOME ONE of the major issues for the survival and development of the world. Since the 21st century, with the expansion of human activities and frequent cross-border exchanges, the spread of emerging infectious diseases in the world has become increasingly serious, posing a threat to human security. Take the coronavirus as an example: there have been three pandemics since the turn of the century: SARS in 2003, Middle East Respiratory Syndrome in 2012, and SARS-CoV-2 in 2019. Biosafety threats have roused widespread concern about biosafety issues and legislation in major countries worldwide. Many countries have unified biosafety into national security systems and have constantly improved biosafety legislation, law enforcement, and supervision systems, responding to the variability and complexity of international and domestic biosafety situations. For example, in 1993, New Zealand's Ministry for Primary Industries formulated the Biosecurity Act,

which aims at ensuring biosecurity and preventing alien invasive species.¹ In 2018, the United States issued the National Biodefense Strategy, which aims to explain how the United States government will manage its activities to more effectively assess, prevent, detect, prepare for, respond to, and recover from biological threats, coordinating its biodefense efforts with those of international partners, industry, academia, non-governmental entities, and the private sector.²

Facing this severe challenge, China is also gradually establishing a national biosafety legal system. Since the 1980s, China has successfully formulated laws and regulations related to biosafety. At present, these laws and regulations have covered seven fields, including: (1) prevention and control of infectious diseases, (2) prevention and control of animal and plant epidemics, (3) prevention of invasive alien species and protection of biological diversity, (4) protection of biological and human genetic resources, (5) laboratory biosafety, (6) regulation on research and development and application of biotechnology, and (7) prevention of biological terrorist attacks and biological weapon threats.³ In September 2018, the

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¹Biosecurity Act 1993, <https://www.legislation.govt.nz/act/public/1993/0095/latest/DLM314623.html>

²National Biodefense Strategy, <https://trumpwhitehouse.archives.gov/wpcontent/uploads/2018/09/National-Biodefense-Strategy.pdf>

³Gao Hucheng, *Explanation of the Draft of Biosafety Law of the People's Republic of China*, 5 BULLETIN OF THE STANDING COMMITTEE OF THE NATIONAL PEOPLE'S CONGRESS 744, (2020).

Standing Committee of the 13th China's National People's Congress (NPC) included the Biosafety Law into the Class-3 legislative projects, which means it requires further research and verification. Then in 2019, the NPC, China's supreme legislature, upgraded the draft of the Biosafety Law for direct deliberation, which accelerates the legislation process. The global SARS-CoV-2 pandemic added a further sense of urgency.⁴ On October 17, 2020, the Biosafety Law was officially voted and then took effect on April 15, 2021, thus finalizing the institutionalization of China's biosafety legislation and marking a major milestone for China's biosafety governance.⁵

The issues of biosecurity legislation have been widely considered by scholars from all over the world. Many domestic and foreign studies introduce and compare the biosafety strategies and legal systems of major countries and regions globally, such as the United States,⁶ the UK, and Australia.⁷ Before the promulgation of the Biosafety Law, some scholars preliminarily conceived the legislative principles and basic systems of the Biosafety Law in light of China's national conditions,⁸ and some scholars paid attention to the legal issues in the subfields of biosafety, such as wildlife protection,⁹ laboratory safety,¹⁰ prevention and control of infectious diseases,¹¹ etc. After the promulgation of the Biosafety Law, some studies have combed the legislative timeline of China's Biosafety Law¹² and the linkage thereof with international and domestic regulations.¹³ However, there are few studies on the legislative model of the Biosafety Law and its prominent role in epidemic prevention and control. At present, the Biosafety Law is the highest level and the latest normative text in the field of biosafety in China. Its legislative orientation and the legislative mode not only changed the strategic planning and governance model in the field of biosafety in China, but also influenced the revision direction of other rules in this field. In addition, China's Biosafety Law was born when the SARS-CoV-2 epidemic spread around the world, making its fundamental role in daily prevention during the epidemic more worthy of discussing.

2. LEGISLATIVE ORIENTATION OF BIOSAFETY LAW

2.1 The Biosafety Law is an integral part of the national security system

The national security system is not a closed system, but a pluralistic and open one. Many countries have made biosafety strategies an important part of

their national security strategies. For example, the National Biodefense Strategy issued by the United States in 2018 is aligned with the 2018 National Security Strategy.¹⁴ In the first meeting of the Central National Security Council in 2014, President Xi Jinping emphasized: "The national security system is an integrated system of political security, homeland security, military security, economic security, cultural security, social security, science and technology security, information security, ecological security, resource security, and nuclear security."¹⁵ Then the National Security Law promulgated in 2015 stipulated in principle the main areas and

⁴Xiaoli Wang, *Enhancing the National Biosecurity System in China amidst COVID-19 Epidemic*, 2 JOURNAL OF BIOSAFETY AND BIOSECURITY 3, (Apr. 17, 2020).

⁵Cong Cao, *China's Evolving Biosafety/Biosecurity Legislations*, 8 JOURNAL OF LAW AND THE BIOSCIENCES 1, (Jun. 30, 2021).

⁶Changmin Liu, Mingjing Song, *American Biological Defense Strategy and National Security*, 3 INTERNATIONAL SECURITY STUDY 96, (2020).

⁷Shilu Zhao, Renjie Cui, *Status Quo, Characteristics and Enlightenment of Australian Biosecurity Legal System*, 4 PORT HEALTH CONTROL 22, (Aug. 30, 2021).

⁸Chaofer Yang, *Basic Principles and Systems of Biosafety Law*, 5 JOURNAL OF BEIJING UNIVERSITY OF AERONAUTICS AND ASTRONAUTICS (SOCIAL SCIENCES EDITION) 22, (2019); Jian Ke, *Principle of Risk Prevention in Legal Protection of Biological Safety*, 3 LAW SCIENCE MAGAZINE 33, (2001).

⁹Xuanlin Liu, *System, Defect and Perfection Path of Wildlife Legal Protection in China*, 8 LAW SCIENCE MAGAZINE 123, (2021).

¹⁰Guizhen Wu, *Laboratory Biosafety in China: Past, Present, and Future*, 1 BIOSAFETY AND HEALTH 56, (Oct. 31, 2019); Yuan Zhiming, *Current Status and Future Challenges of High-level Biosafety Laboratories in China*, 1 JOURNAL OF BIOSAFETY AND BIOSECURITY 123, (Oct. 24, 2019).

¹¹Tauseef Ahmad, et al., *Biosafety and Biosecurity Approaches to Restrict/Contain and Counter SARS-CoV-2/COVID-19 Pandemic: A Rapid-Review*, 44 TURKISH JOURNAL OF BIOLOGY 132, (Jun. 21, 2020).

¹²Cong Cao, *China's Evolving Biosafety/Biosecurity Legislations*, 8 JOURNAL OF LAW AND THE BIOSCIENCES 1, (Jun. 30, 2021).

¹³See Weiwei Han, *Introduction on Biosafety Law of the People's Republic of China and Linkage Thereof with International and Domestic Regulations*, 40 BIOTECHNOLOGY LAW REPORT 95, (2021).

¹⁴National Biodefense Strategy, <https://trumpwhitehouse.archives.gov/wpcontent/uploads/2018/09/National-Biodefense-Strategy.pdf>

¹⁵"Adhere to the Overall National Security Concept and Take the National Security Road with Chinese Characteristics," *Xinhuanet*, (Apr. 15, 2014), http://www.xinhuanet.com/politics/2014-04/15/c_1110253910.htm

overall objectives of national security, which laid the foundation for improving the biosafety legal system. On February 14, 2020, at the 12th meeting of the Central Committee for Comprehensively Deepening Reform, President Xi declared again that biosafety should be brought into the national security system.¹⁶ Therefore, the newly promulgated Biosafety Law and the National Security Law have the relationship between “general provisions” and “specific provisions,”¹⁷ which fill the gap in the national security system in the field of biosecurity.

2.2 Biosafety Law is the basic and comprehensive legislation in the field of biosafety

Suppose the national security legal system is regarded as a whole under the guidance of the National Security Law. In that case, the biosafety legal system is one of the branches, and the Biosafety Law is core of that branch.¹⁸ Although the legislative process of the Biosafety Law has been delayed until 2019, China has issued many regulations and guidelines on specific areas of biosecurity in recent years. The Biosafety Law is a positive development insofar as the law fills gaps where the existing statutes may have been too general, and so may guide the available regulations and constitute a systematic biosafety legal system. At present, the Biosafety Law consists of 10 chapters and 88 articles, which are mainly divided into eight regulatory areas, such as prevention and control of major emerging infectious diseases, animal and plant epidemics, biotechnology research, development and application, biosafety management of pathogenic microorganism laboratories, safety management of human genetic resources and biological resources, prevention of alien species invasion, and protection of biodiversity. The newly promulgated Biosafety Law is like a central nerve that connects all the relevant laws and regulations in these eight regulatory areas, which can comprehensively coordinate scattered system norms in various fields and avoid conflicts in the operation of various laws and regulations (See Table 1).

3. LEGISLATIVE MODEL OF BIOSAFETY LAW

3.1 Biosafety law adopts a policy-oriented legislative model

Environmental legislation can be divided into two modes: policy-oriented legislation and regulatory legislation,¹⁹ and this classification is also applicable to legislation in the field of biosafety. *Regulatory legislation* focuses on determining the specific regula-

tory structure, which requires clear authority(ies) and normative objects, clear management rules, and legal responsibilities. Therefore, the regulatory legislative mode is more suitable for legislation in a specific field. In contrast, *policy-oriented legislation* does not focus on particular control items, specific control tools and implementation means, but rather aims at clarifying policy orientation, basic principles, and basic systems. Therefore, China's Biosafety Law adopted the policy-oriented legislative model as biosafety governance involves fields at different levels, which inevitably requires a multi-sectoral collaborative management mechanism and diversified regulatory methods. On the one hand, the Biosafety Law has established a relatively complete legal framework and defined the basic governance objectives, competent departments, basic systems and procedures in different fields, which can effectively guide the relevant departments to establish more detailed systems or make the existing systems operate within the legal framework. On the other hand, adopting a policy-oriented legislation mode can avoid the defects when a piece of legislation governs multiple regulatory fields, such as the logical incoherence of internal legal provisions and the length and redundancy of the legislation. The provisions of the Biosafety Law contain some guiding clauses that can be connected with corresponding special legislations, which not only avoid the risk of repeated legislation, but also provide a fundamental basis for the implementation of these special legislations.

3.2 The Biosafety Law takes risk prevention as the basic principle

By analyzing the previous legislative provisions, it can be found that China's biosafety governance focuses on emergency remedy and recovery after damage, but lacks monitoring and prevention

¹⁶“Perfecting the System and Mechanism of Major Epidemic Prevention and Control and Perfecting the National Public Health Emergency Management System,” *People's Daily*, (Feb. 15, 2020), <http://cpc.people.com.cn/n1/2020/0215/c64094-31588184.html>

¹⁷Jiaxiang Hu, *Legislative Orientation and Legal Application of China's Biosafety Law—From the Perspective of Genetically Modified Food Regulation*, 20 *FRONTIERS* 22, (2020).

¹⁸Tianbao Qin, *Legislative Orientation and Development of Biosafety Law*, 3 *SOCIAL SCIENCE JOURNAL* 134, (2020).

¹⁹Junrong Ye, *Two Models of Environmental Legislation: Policy Legislation and Regulatory Legislation*, *THE BALANCE OF RULE OF LAW IN TSINGHUA SERIES 19: ENVIRONMENTAL RULE OF LAW AND CIVILIZATION TRANSFORMATION*, Tsinghua University Press, 2013, p. 6.

TABLE 1. BIOSAFETY LAW AND DOMESTIC LAWS AND REGULATIONS

<i>Relevant Chapters of Biosafety Law</i>	<i>Primary Coverage</i>	<i>A List of Relevant Current Laws and Regulations</i>
Chapter 3	<i>Prevention and Control of Major Emerging Infectious Diseases, Animal and Plant Epidemics</i>	Infectious Diseases Prevention and Control Law, Animal Epidemic Prevention Law, Emergency Response Law, Livestock and Poultry Epidemic Prevention Regulations, Emergency Regulations for Public Health Emergencies, Interim Measures for the Management of Medical Laboratories, Guide to Laboratory Biosafety in novel coronavirus, Customs Law, Animal and Plant Import and Export Law, Import and Export Commodity Inspection Law, Frontier Health Quarantine Law, Entry and Exit Animal and Plant Quarantine Law, Regulations on Import and Export Administration of Goods, Regulations on Plant Quarantine, Notice of the National Health and Health Commission on Further Strengthening the Management of Antimicrobial Drugs to Cure Drug Resistance
Chapter 4	<i>Research, Development and Application of Biotechnology</i>	Regulations on Safety Management of Agricultural Genetically Modified Organisms (revised in 2017), Measures for Safety Management of Biotechnology Research and Development, Measures for Management of Clinical Application of Medical Technology, Regulations on Safety Management of Biotechnology Research and Development (Draft for Comment), Regulations on the Administration of Clinical Application of New Biomedical Technologies (Draft for Comment), Administrative Measures for Safety Evaluation of Agricultural Genetically Modified Organisms, Guidelines for Safety Evaluation of Agricultural Genetically Modified Organisms (Plants, Animals and Animals Microbes), Administrative Measures for Labeling of Agricultural Genetically Modified Organisms, Measures for Approval of Agricultural Genetically Modified Organisms Processing, Measures for Safety Management of Agricultural Genetically Modified Organisms Import, Measures for Inspection and Quarantine Management of Inbound and Outbound Genetically Modified Organisms, General Requirements Laboratory for Safety Management of Agricultural Genetically Modified Organisms, Labeling of Agricultural Genetically Modified Organisms
Chapter 5	<i>Pathogenic Microorganisms, Laboratory Biosafety</i>	Regulations on Biosafety Management of Pathogenic Microbiology Laboratories, Measures for Biosafety Environmental Management of Pathogenic Microbiology Laboratories, Measures for Examination of Construction of High-grade Pathogenic Microbiology Laboratories, Measures for the Administration of Biosafety Examination and Approval of Laboratory and Experimental Activities of Highly Pathogenic Microbiology Infected by Human Beings, Measures for the Administration of Human-to-Human Infectious Pathogenic Microbiological Bacteria (Viruses) Depository Institutions, Regulations for the Administration of Transportation of Highly Pathogenic Microbiological Bacteria (Viruses) or Samples that can infect humans, Classification List of Animal Pathogenic Microbes, Measures for the Administration of Clinical Laboratories of Medical Institutions.

(continued)

TABLE 1. CONTINUED

<i>Relevant Chapters of Biosafety Law</i>	<i>Primary Coverage</i>	<i>A List of Relevant Current Laws and Regulations</i>
Chapter 6	<i>Human Genetic Resources and Biological Resources Security</i>	Regulations on Management of Human Genetic Resources (revised in 2019), Safety Management Measures of Genetic Engineering, Management Measures of Human Assisted Reproductive Technology, Technical Specifications of Human Assisted Reproductive Technology, Ethical Principles of Human Assisted Reproductive Technology and Human Sperm Bank, Administrative Measures for Clinical Research of Stem Cells, Ethical Review Measures for Biomedical Research Involving People, Seed Law, Wildlife Protection Law, Nature Reserve Regulations, Wildlife Protection Regulations, Land Wildlife Protection Regulations, Livestock and Poultry Breeding Management Regulations and New Plant Variety Protection Regulations
Chapter 7	<i>Guard against Bioterrorism and Threats of Biological Weapons</i>	
Chapter 8	<i>Biosafety Capacity Building</i>	Food Safety Law, Agricultural Product Quality Safety Law, Drug Safety Law and Regulations on Cosmetics Supervision and Administration

systems to address potential, but not actualized, risks.²⁰ For example, in the 2013 Amendment to the Law on the Prevention and Control of Infectious Diseases, although the basic principle of “prevention first” had been established, Chapter 2 “Prevention of Infectious Diseases” still lacks the critical risk-prevention systems such as risk-judgment system and risk-assessment system for infectious diseases. This reflects that the principle of risk prevention has not yet penetrated through the biosafety laws and regulations system. Therefore, in many cases, we can only be driven by unexpected events to mitigate or remedy losses after they occur instead of eliminating the risks as soon as they appear. However, risk prevention and control mechanisms are indispensable because of the seriousness, irreversibility, and extensive influence of biosafety risks and their enormous potential damage consequences. Furthermore, establishing and implementing the principle of risk prevention in biosafety legislation is an obligatory requirement of China as a party to the Convention on Biological Diversity, Cartagena Protocol on Biosafety, and other international conventions, which have been gradually adopted by countries all over the world.²¹ The outbreak of the SARS-CoV-2 epidemic highlights the importance of the principle of risk prevention in national security legislation again. Drawing lessons from the legislative spirit embodied in international conventions, the importance of risk prevention has been emphasized in the legislative text of the Biosafety Law. According to Articles 1 and 3 of the Biosafety Law, the “prevention of bio-

safety risks” is not only one of the legislative purposes, but also the basic principle adhered to by this law. In addition, the Biosafety Law has also established a series of national systems for biosafety risk prevention, covering 11 biosafety-related aspects from monitoring and early warning; risk investigation and assessment; information sharing; information disclosure; catalogs and lists; standards; reviews; emergency responses; investigation and source tracing for incidents; entry approval; through to responses to major overseas incidents, which has promoted the normalization of biosafety risk prevention in China and also provided a legal scheme for the containment of SARS-CoV-2 pandemic.

4. POSITIVE SIGNIFICANCE FOR THE SARS-COV-2 PREVENTION AND CONTROL

The outbreak of SARS-CoV-2 at the end of 2019 has increased, as never before, the vigilance of

²⁰Bingqing Lu, *Paradigm Transformation of Biosafety Legal Regulation—From Damage Management to Risk Management*, 1 JOURNAL OF JISHOU UNIVERSITY (SOCIAL SCIENCES) 34, (2021).

²¹Kang Wang, et al., *The Legal System of National Biosafety with Chinese Characteristics: The Status Quo and Trends*, 2021 4TH INTERNATIONAL WORKSHOP ON ADVANCES IN SOCIAL SCIENCES (IWASS 2021), Francis Academic Press, pp. 122–127.

Chinese society against major infectious diseases and the government's focus on national biosecurity. As of when this article was first begun, the SARS-CoV-2 had infected 101,683 people in China and caused 4,636 deaths,²² and has severely impacted China's economic development (6.1% GDP growth in 2019 to 2.3% in 2020). Although China, with its strong political means and institutional strengths, has rapidly contained the transmission of the SARS-CoV-2,²³ at the same time, the global epidemic situation is particularly challenging. As mentioned above, the world has *at least* 276.5 million confirmed cases with 5,403,662 deaths.²⁴ Even more worryingly, SARS-CoV-2 has evolved to sets of mutation that have been proved to reduce the neutralization of the postvaccination serum by emerging evidence,²⁵ such as the Omicron and Delta variants. And the forces propelling the virus to evolve to evade immunity are likely to grow stronger as most of the planet gains immunity to the virus through infection, vaccination, or both.²⁶ In this context, Chinese society and the authorities are in urgent need of a law that can help counteract the SARS-CoV-2, as well as prevent and deal with the potential major epidemic.

As mentioned above, although China has a series of laws and regulations in the field of biosecurity, these laws and regulations did not play the role of early warning, guidance, and regulation as they should have done in the early stages of the outbreak of the SARS-CoV-2, but instead exposed that the nation's existing legal system of biosecurity was full of loopholes. To be specific, the prevention mechanism of major infectious diseases in China has serious legislative loopholes in early warning mechanism, rapid response mechanism in accordance with the law, and the division of government functions. The current Biosafety Law establishes a special chapter for the prevention and control of major outbreaks of emerging infectious diseases, animal and plant epidemics, focusing on the improvement of these three institutional weaknesses, in an attempt to enhance China's prevention and control capacity for major infectious diseases. In short, the Biosafety Law will have an important positive effect on China's current and future epidemic prevention and control.

4.1 Biosafety Law has enhanced China's early warning capability for infectious diseases

The outbreak of SARS-CoV-2 exposed the lagging early warning mechanism of the epidemic in China. It is widely believed in the medical profession that the epidemic started in the middle of

December 2019 in China,²⁷ but there was no official acknowledgment or full fighting from the government until January 20, leaving around 2,700 laboratory-confirmed infections including 80 deaths in the month.²⁸ The slow response was not due to poor medical technology or capacity to identify this highly threatening coronary infection. On the contrary, Chinese doctors in Wuhan, *e.g.*, Dr. Zhang Jixian, had discovered and reported the "abnormal pneumonia of unknown cause" as early as December 27,²⁹ but this report didn't get the attention it deserved from the Center for Disaster Control (CDC), the core authority for epidemic prevention and control in China.

In response to this lesson, the newly enacted Biosafety Law builds an early warning system for major and sudden outbreaks of infectious diseases at three levels: the state level, professional institution level, and individual level. Firstly, at the national level, the Biosafety Law requires the State Council to organize relevant departments to layout surveillance stations for emerging infectious diseases, establish a surveillance network, improve the surveillance information reporting system, and take the initiative to carry out surveillance and pathogen testing (Article 27). These specific

²²"The Latest COVID-19 Situation as of 24:00 On 28 December," *China's Health Committee* (Dec. 28, 2021), http://www.gov.cn/xinwen/2021-12/29/content_5665094.htm.

²³Zi Zhengfa and Wu Wanqiang, *Amendment of China's Biotechnology Laws in Relation to the Prevention and Containment of the SARS-COV-2 Pandemic*, 39 *BIOTECHNOLOGY LAW REPORT* 6, 458 (Dec. 2, 2020).

²⁴"WHO Coronavirus Disease (COVID-19) Dashboard," *World Health Organization* (Dec. 17, 2021), <https://covid19.who.int/>

²⁵William T. Harvey, et al., *SARS-CoV-2 Variants, Spike Mutations and Immune Escape*, *NAT REV MICROBIOL* 19, 409c (Jun. 1, 2021).

²⁶Ewen Callaway, *Beyond Omicron: What's Next for COVID's Viral Evolution*, *NATURE* (Dec. 7, 2021).

²⁷Qun Li, et al., *Early Transmission Dynamics in Wuhan, China, of Novel Coronavirus-Infected Pneumonia*, 382 *NEW ENGLAND JOURNAL OF MEDICINE*, 1199, 1207 (Mar. 26, 2020).

²⁸Peng Zhou, et al., *A Pneumonia Outbreak Associated with a New Coronavirus of Probable Bat Origin*, *NATURE* 579, 270 (Feb. 3, 2020).

²⁹It is worth noting that, thanks to Dr. Zhang Jixian's professional medical knowledge and quick emergency measures, no one in her department contracted COVID-19. See "Hubei Province Gives Great Credit to the First Person Who Sounded the Alarm for the Outbreak of SARS-CoV-2," *China News* (Feb. 6, 2020), <https://www.chinanews.com.cn/sh/2020/02-06/9081381.shtml>

departments mainly include the National Health Commission, the Ministry of Agriculture and Rural Affairs, the National Forestry and Grassland Administration, the General Administration of Customs, and the Ministry of Ecology and Environment. At the level of professional bodies, the Biosafety Law requires CDCs to carry out proactive surveillance, collection, analysis, and reporting of known infectious diseases and unexplained diseases included in the scope of surveillance, and to predict potential emerging outbreaks of infectious diseases. In addition to this, the Biosafety Law learns from the case of Dr. Zhang Jixian that medical institutions and CDCs are required to take protective measures in a timely manner when an unexplained disease is detected and reported. Finally, the Biosafety Law requires that all individuals who discover an infectious disease should promptly report it to medical institutions, CDCs, and other relevant departments.

The Biosafety Law aims to strengthen China's early warning capability for major infectious diseases and avoid another sudden outbreak by building a three-in-one monitoring and warning system of "state-professional institutions-individuals."

4.2 The Biosafety Law has enhanced China's capacity to fight the epidemic in accordance with the law

Although the Chinese government was able to contain the spread of the SARS-CoV-2 in a very short period, it is found that China's rapid prevention and control of the epidemic relied mainly on political, rather than legal means. It took China's government roughly two months to go from a daily increase of 15,000 cases to a daily increase of single digits; however, many timely and effective epidemic control measures were initiated by political force rather than by law, and some management measures and measures to block the spread of the epidemic are suspected of overstepping the law.³⁰ China's President Xi Jinping specifically stressed that "the more the epidemic is prevented and controlled, the more we must adhere to the law." Finally, authorities also attributed the success of the fight against the SARS-CoV-2 to the strength of the political system rather than the legal system.³¹

In emergency situations, timely and effective administrative actions are essential to protect the lives and health of citizens, but as a country governed by the rule of law, it is equally important for China to build relevant legal systems to keep pace with the development of the actual situation. The Biosafety Law, therefore, provides detailed provisions for a

rapid response mechanism in the event of an outbreak of major epidemic (Article 15 and Article 21).

Firstly, the law requires the establishment of a biosafety risk investigation and assessment system at the national level, which empowers the relevant authorities to take the necessary risk prevention and control measures in a timely manner when a potential biosafety risk is identified, when a relevant report is received, or when a major epidemic situation is faced. Secondly, the law requires the establishment of a biosecurity emergency response system at the national level, with relevant departments of the State Council establishing relevant biosecurity emergency plans by sectors and fields. All governments at the county level and above are required to develop local biosecurity emergency plans, prepare them in advance, train personnel, and conduct drills.

The investigation and assessment systems and the emergency response system have greatly enhanced the Chinese government's ability to take rapid measures against infectious diseases following the law.

4.3 The Biosafety Law has clarified the government's division of function

Another problem exposed in China's epidemic prevention and control is the lack of a clear division of responsibilities for launching epidemic prevention measures. The conflict between the governing principles of the Emergency Response Law and the Law on Prevention and Control of Infectious Diseases, and the division of authority of the epidemic prevention headquarters, are in urgent need of resolution at the legal level.³²

The Biosafety Law establishes the basic principle of "national guidance, local enforcement" basic principles, with greater emphasis on the principle of territorial jurisdiction (Article 10). The benefits can be guided by the unity of the country's most authoritative organization, to carry on the emergency

³⁰Zi Zhengfa and Wu Wanqiang, *Amendment of China's Biotechnology Laws in Relation to the Prevention and Containment of the SARS-CoV-2 Pandemic*, 39 BIOTECHNOLOGY LAW REPORT 6, 464 (Dec. 2, 2020).

³¹"The Nationwide Fighting against SARS-CoV-2 Demonstrates the Strength of China's Political System," *Guangming Daily* (Jul. 10, 2020), http://www.xinhuanet.com/politics/2020-07/10/c_1126219676.htm

³²Zi Zhengfa and Wu Wanqiang, *Amendment of China's Biotechnology Laws in Relation to the Prevention and Containment of the SARS-CoV-2 Pandemic*, 39 BIOTECHNOLOGY LAW REPORT 6, 464 (Dec. 2, 2020).

duties demanded of the government by means of local government(s), according to the actual situation, dynamic deployment, and epidemic prevention measures. This principle has the advantage of imposing emergency responsibility on the local governments while letting the most authoritative authority to provide unified guidance.

Under this epidemic prevention system of “national guidance, local enforcement,” the Biosafety Law further clarifies and delineates the responsibilities of specific departments. As mentioned above, China’s biosecurity is located under the national security system, as the Central National Security Agency is at the top of the epidemic prevention system in the Biosecurity Law (Article 5 of the National Security Law), and is responsible for setting up the coordination mechanism to ensure that the top-level design of the mechanism is scientific and that the most resources are mobilized in the most effective way. The coordination mechanism for specific epidemic prevention work is made up of the State Council’s health department, the Ministry of Agriculture and Rural Affairs, the Ministry of Science and Technology, the Ministry of Foreign

Affairs, and several military authorities. According to their division of responsibilities, these specialized state organs will be responsible for specific epidemic prevention work. In addition, this coordination mechanism, composed of various professional departments, will also establish an expert committee, which will stand outside the state authorities and provide technical support for the epidemic prevention work from a professional perspective, such as consultation, evaluation and validation (Article 12). Finally, the Biosafety Law explicitly requires local governments to be responsible for epidemic prevention work within their administrative regions, especially governments at the county level and above, which will organize specific departments to carry out detailed epidemic prevention work (Article 13). Grassroots self-government organizations, such as village and neighborhood committees, are also explicitly required to assist local governments in carrying out relevant epidemic prevention work, while units and individuals should cooperate.

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